Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. TEAM AND PROJECT OVERVIEW

Team Name

Holdwise

Selected Domain

Domain 2: Orbyte Smart SME Challenge (Alpha Red Solutions)

Selected SDG Track (1–17)

SDG 8: Decent Work and Economic Growth

Project / Solution Name

Holdwise: WhatsApp enquiry-to-quotation module for Orbyte

Name (Leader)

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University Affiliation(s)

Xiamen University Malaysia (XMUM)

Number of Team Members

2

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysia has 1,101,725 micro, small and medium enterprises: 96.9% of all businesses, and 69.7% of them are micro firms [1]. Many of them, such as hardware shops, printers and caterers, take orders through WhatsApp messages like "boss, nak 20 bag simen + 5 batang paip PVC 2 inci, hantar Bangi esok, berapa?".

Today the owner reads every message, decodes mixed Malay and English shorthand, looks up prices in a notebook or Excel, checks stock, adds delivery and types a reply. One quote takes several minutes of skilled time, replies arrive hours later or not at all, and a wrong price or an out-of-stock item is easy to miss.

Speed decides the sale: firms that answer an enquiry within an hour are nearly seven times as likely to qualify the lead as firms that wait longer, yet only 37% of firms manage it and 23% never answer [2].

Those affected are micro-business owners and their few staff, who spend evenings answering messages instead of growing the business, and their customers, contractors and households who wait for a price. The issue is economic (lost revenue, low productivity, pricing mistakes) and social (owner burnout, small shops losing customers to larger competitors with sales teams). Existing chatbots answer FAQs, but they do not produce a priced, stock-checked quotation that a shop owner can trust.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Holdwise is a quotation module for the Orbyte SME hub. It turns a WhatsApp price enquiry into a ready-to-send quotation in under a minute, while a person stays in control. It is a five-step pipeline, not a chatbot:

1. Trigger: an enquiry arrives in the Inbox.

2. Data: catalogue, prices, stock, customer type and delivery zones are loaded from the database.

3. Decision: GenAI does one job only. GLM-5.3, through the sponsor's RunPod endpoint, reads the messy Malay/English/Manglish text and returns strict JSON (items, quantities, units, confidence, delivery area), validated with a Zod schema. The model never sees prices. Deterministic rules then match items to the catalogue, price them, apply discounts and delivery fees and check stock. Anything uncertain (unknown item, ambiguous variant, confidence below 0.7, missing quantity, low stock) is flagged with a plain-language reason. If the AI is switched off, takes over 8 seconds or returns invalid JSON, a rules parser takes over and the draft says which one was used.

4. Approval: staff review and edit the draft; quotes above RM 500, discounts above 5% or price overrides go to the Owner.

5. Outcome: the approved quote is sent (simulated WhatsApp), stock is reserved, one Orbyte credit is charged and every action is written to an audit log. A dashboard shows quotes sent, response time against the manual baseline, hours saved and errors caught.

Workshop skills we use: GenAI APIs with structured outputs (GLM-5.3, JSON schema validation and an evaluation set comparing AI with rules), database design (PostgreSQL with Drizzle ORM for products, customers, quotes and the activity log) and cloud deployment (Replit Deployments with secrets management, live at quote-lah-hardware-quotation-mvp.replit.app).

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: Next.js 15 (App Router, static export), React 19, TypeScript, Tailwind CSS 4, typed TanStack Query clients generated from an OpenAPI spec (Orval).

Backend: Node.js 24 with an Express 5 REST API; Zod validates every request and every AI answer.

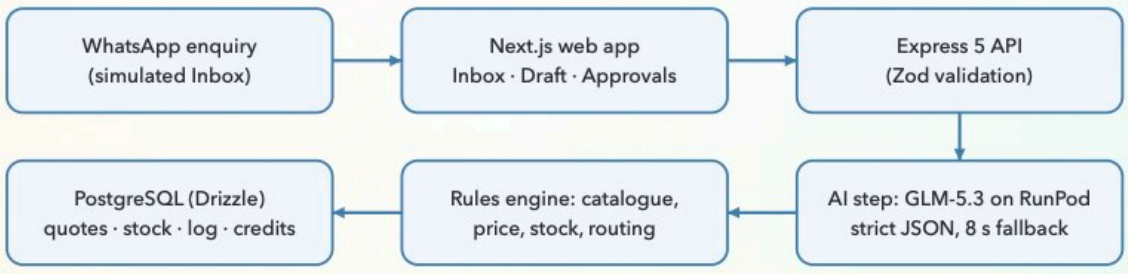
Database: PostgreSQL on Replit with Drizzle ORM (products, customers, quotes, stock reservations, activity log, credit wallet). All data is synthetic.

AI: GLM-5.3 on the RunPod Public Endpoint (OpenAI-compatible) for message extraction only; a TypeScript rules engine does matching, pricing, stock checks and approval routing; a rules parser is the fallback.

Hosting: Replit Deployments (Autoscale) with Replit Secrets; code on GitHub.

Sponsor resources: Yes. RunPod: the GLM-5.3 Public Endpoint, paid from the US$25 team credit, reads each message into JSON (reasoning effort low, at most 1,000 tokens per call). Replit: Replit Agent for the first scaffold, plus Deployments, PostgreSQL and Secrets for the live prototype.

System block diagram:



3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: a team member had seen a relative's hardware shop answer price requests on WhatsApp by hand, so we collected the typical message styles: Malay, English, Manglish, abbreviations and missing quantities.

Define: the pain is not chatting but the slow, error-prone step from message to priced quote, and the owner must keep the final say.

Ideate: we rejected a chatbot and a fully autonomous agent. AI only reads the text; rules handle money and stock; a human approves every quote.

Prototype and test: we built the full pipeline on synthetic data in Replit, drafted every seed message and fixed what broke (decimal points, words like "harga", place names), then added confidence flags with plain-language reasons. A 20-message evaluation set with 5 held-out messages measures AI against rules: GLM-5.3 read 37/37 items, the rules 34/37.

Deployment readiness: the app is live on Replit Deployments with PostgreSQL and secrets in Replit Secrets. It keeps working without AI, has role-based approval and an audit log, and is shaped as an Orbyte module, so Phase 2 can connect real messaging and inventory APIs.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8.2: higher economic productivity through technological upgrading and innovation, focusing on labour-intensive sectors. SDG 8.3: support the growth of micro, small and medium-sized enterprises. Holdwise automates the most labour-intensive sales task of a micro retailer, quoting, while people stay in control.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Manual quoting takes an assumed 8 minutes of owner or staff time per WhatsApp enquiry (team estimate, editable in the app; to be validated with shop owners in Phase 2), and replies often take hours: only 37% of firms answer within an hour and 23% never do [2]. Without AI, our rules-only parser reads 34/37 items (91.9%) on our synthetic test set, but only 5/8 (62.5%) on unseen messages [3]. Scale: 1,101,725 MSMEs, 96.9% of establishments, 69.7% micro [1]. Sources: [1] SME Corp. Malaysia, MSME Insights 2023/24 (DOSM data). [2] J. Oldroyd, K. McElheran, D. Elkington, "The Short Life of Online Sales Leads", Harvard Business Review 89(3), 2011. [3] Holdwise evaluation, data/eval-set.json, 24 Oct 2026.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	1. Staff handling time per quote: minutes from message received to quote sent, logged automatically in the activity log. 2. Item extraction accuracy on unseen messages: % of requested items matched to the right catalogue SKU and quantity. 3. Pricing or stock errors that reach customers: count per 100 quotes.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	1. At least 75% less staff time per quote (from 8 minutes to 2 minutes or less, median). 2. At least 95% item accuracy on a held-out set of 50 or more realistic messages (today: GLM-5.3 8/8 on our small held-out set, rules 62.5%). 3. Zero unapproved or mispriced quotes sent: 100% of quotes approved by a person.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☑ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Yerassyl Mubarak earlier built a simple price-reply automation for a relative's shop. None of its code, design or data is used here: Holdwise was written from scratch during the hackathon, in a repository created after the start date. That experience only helped us understand the problem.

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
GLM-5.3 (zai-org) via RunPod Public Endpoint	Model: pre-trained LLM called through an API (no fine-tuning)	RunPod Terms of Service; Z.ai model terms
Next.js 15, React 19, Tailwind CSS 4, Radix UI, TanStack Query, lucide-react	Code: web frameworks and UI libraries	MIT (lucide-react: ISC)
Express 5, Zod 3, Orval	Code: API server, validation, client generator	MIT
Drizzle ORM, node-postgres, PostgreSQL	Code: database access and database engine	Apache-2.0; MIT; PostgreSQL License
Google Gemini API (optional backup, not active in the deployed app)	Model: pre-trained LLM called through an API	Gemini API Terms of Service

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

None.

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



Yerassyl Mubarak

[Date]

24 October 2026